Lecture 15: Introduction to Nonlinear Optimization

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Agenda

- Nonlinear optimization (about 4 weeks)
- Integer optimization (about 1.5 weeks)
- Homework 6 posted, due on Apr 12th at noon
- Lecture 16 will be conducted online

Introduction to Nonlinear Optimization

So far we have discussed linear optimization problems. However, in practice, there are many interesting optimization problems that do not take a linear form.

In general, we can write a nonlinear optimization problem as:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in F \end{aligned}$$

We call F the feasible region and $\mathbf{x} \in F$ feasible solutions.

In the following, we study such nonlinear optimization problems.

- The properties of such problems.
- How to find the optimal solution?
- Without otherwise specified, we always assume we are solving a minimization problem.

Global and Local Optimizers

We call $\bar{\mathbf{x}}$ a *global optimizer (minimizer)* of the optimization problem:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in F \end{aligned}$$

if for all $\mathbf{x} \in F$, $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$.

- The solution we obtained for LP is a global optimizer
- We always want to find global optimizers. However, sometimes this is not easy, we may have to settle on *local optimizers*

Local Minimizers

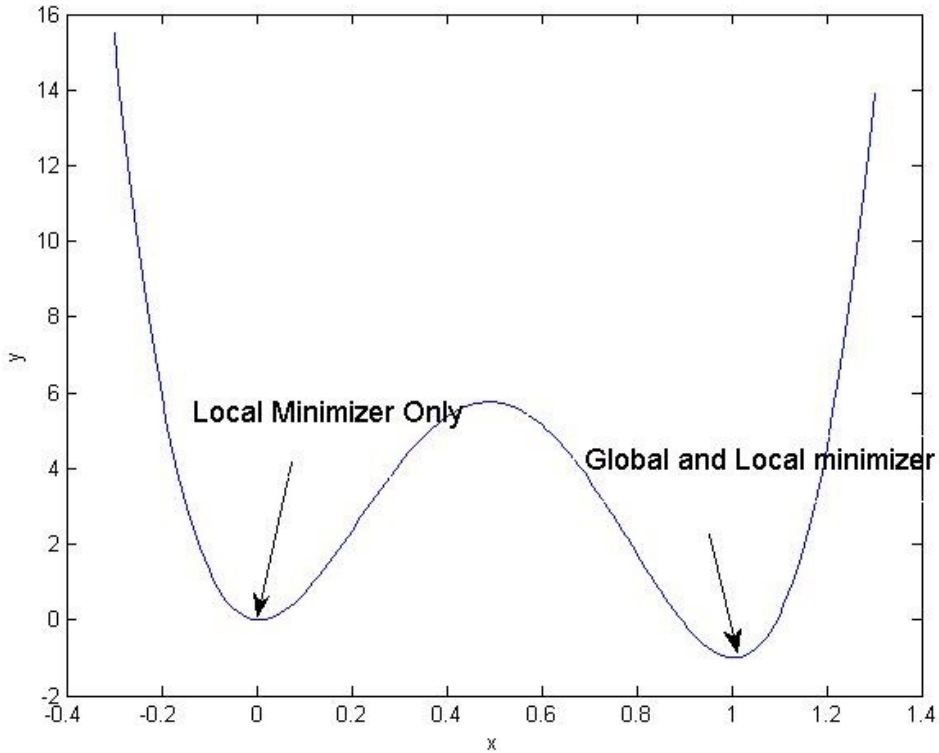
We call $\bar{\mathbf{x}}$ a *local optimizer (minimizer)* of:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in F \end{aligned}$$

if there exists a neighborhood $N(\bar{\mathbf{x}})$ of $\bar{\mathbf{x}}$ (a small ball around $\bar{\mathbf{x}}$) such that for all $\mathbf{x} \in N(\bar{\mathbf{x}}) \bigcap F$, $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$.

- Global minimizer is always local minimizer, however, the reverse is not true

Example: $f(x) = 100x^2(1-x)^2 - x$



Review: Gradient, Hessian Matrix and Taylor Expansion

Let $f : \mathbb{R}^n \to \mathbb{R}$.

- Assume $f(\mathbf{x}) = f(x_1, x_2, ..., x_n)$ is continuously differentiable. Then we denote the gradient of f by (an $n \times 1$ vector)

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}; \frac{\partial f}{\partial x_2}; ...; \frac{\partial f}{\partial x_n} \right)$$

  By Taylor expansion, we have

$$f(\mathbf{x} + \alpha \mathbf{d}) = f(\mathbf{x}) + \alpha \nabla f(\mathbf{x})^T \mathbf{d} + o(\alpha)$$

- Assume f is second-order differentiable. Then we denote the Hessian matrix of f by (an $n \times n$ matrix)

$$\nabla^2 f(\mathbf{x}) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{i,j}$$

  By Taylor expansion, we have

$$f(\mathbf{x} + \alpha \mathbf{d}) = f(\mathbf{x}) + \alpha \nabla f(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \alpha^2 \mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} + o(\alpha^2)$$

Example

Suppose

$$f(x_1, x_2, x_3) = x_1^2 + x_1 x_2 + x_1 e^{x_3} + x_2 \log x_3$$

Then

$$\nabla f(\mathbf{x}) = \left(2x_1 + x_2 + e^{x_3}; x_1 + \log x_3; x_1 e^{x_3} + \frac{x_2}{x_3}\right)$$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 2 & 1 & e^{x_3} \\ 1 & 0 & \frac{1}{x_3} \\ e^{x_3} & \frac{1}{x_3} & x_1 e^{x_3} - \frac{x_2}{x_3^2} \end{bmatrix}$$

$\downarrow$

$\frac{\partial \nabla f(x)}{\partial x_1}$

Optimality Conditions

In the following, we first study what conditions an optimal solution has to satisfy for nonlinear optimization problems

- Optimality conditions
- We will start with local optimal solutions

Optimality Conditions: Unconstrained Problems

Let's start from the easiest case in which $F = \mathbb{R}^n$ (unconstrained problems).

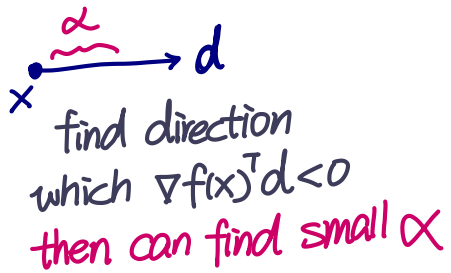
What are the optimality conditions for local minimizers for unconstrained problems?

- Claim: We must have

$$\nabla f(\mathbf{x}) = \underset{\sim}{0} \quad \text{vector} \in \mathbb{R}^n$$

Reason: If $\nabla f(\mathbf{x}) \neq 0$, then we can find a vector $\mathbf{d}$ such that $\nabla f(\mathbf{x})^T \mathbf{d} < 0$. Therefore by Taylor expansion

$$f(\mathbf{x} + \alpha \mathbf{d}) = f(\mathbf{x}) + \alpha \nabla f(\mathbf{x})^T \mathbf{d} + o(\alpha)$$



By choosing α small enough, we can find a point $\mathbf{x}' = \mathbf{x} + \alpha \mathbf{d}$ in the neighborhood of $\mathbf{x}$ such that $f(\mathbf{x}') < f(\mathbf{x})$.

First-Order Necessary Condition (FONC)

Theorem (First-Order Necessary Condition)

If $\mathbf{x}^*$ is a local minimizer of $f(\cdot)$ for the unconstrained problem, then we must have $\nabla f(\mathbf{x}^*) = 0$.

Remark

First-order necessary condition provides all the candidates for local minimizers.

Example: $f(\mathbf{x}) = x_1^2 - x_1 x_2 + x_2^2 - 3x_2$.

The FONC is

$$2x_1 - x_2 = 0, \quad -x_1 + 2x_2 = 3$$

There is a unique solution ($x_1 = 1$, $x_2 = 2$), which turns out to be the global minimizer for f.

Another Example: Least Squares Problem

Assume a variable y is affected by n factors $x_1, ..., x_n$. We know that they approximately have a linear relationship:

$$y \approx \beta_1 x_1 + \cdots + \beta_n x_n$$

Now we want to find out this relationship (parameters βs).

- We have m observations ($m > n$):

$$\{\mathbf{x}_i, y_i\} = \{(x_{i1}, ..., x_{in}), y_i\}, i = 1, ..., m$$

Ideally, we want to find $\boldsymbol{\beta} = (\beta_1, ..., \beta_n)$ such that $y_i = \mathbf{x}_i^T \boldsymbol{\beta}$. Or equivalently, $\mathbf{y} = X\boldsymbol{\beta}$ where X is a matrix whose ij-th entry is x_{ij}

- However, this may not be feasible
- Usually the observations do not follow $y_i = \mathbf{x}_i^T \boldsymbol{\beta}$ exactly. There are noises in the observations.

Least Squares Problem Continued

Instead, we try to minimize the sum of the squared errors

$$\text{minimize}_{\boldsymbol{\beta}} \quad \sum_{i=1}^{m} \left(y_i - \sum_{j=1}^{n} \beta_j x_{ij} \right)^2$$

The matrix form of this problem is

$$\text{minimize}_{\boldsymbol{\beta}} \quad ||X\boldsymbol{\beta} - \mathbf{y}||_2^2 = \boldsymbol{\beta}^T X^T X \boldsymbol{\beta} - 2\boldsymbol{\beta}^T X^T \mathbf{y} + \mathbf{y}^T \mathbf{y}$$

where $||\mathbf{w}||_2^2 = \mathbf{w}^T \mathbf{w} = w_1^2 + \cdots + w_n^2$.

$f(x) = x^T M x = \sum_{ij} x_i x_j m_{ij}$

$\frac{\partial f(x)}{\partial x_i} = \sum_j x_j m_{ij} + \sum_i x_i m_{ij} = 2\sum_j x_j m_{ij} = 2Mx.$

Facts:

- If $f(\mathbf{x}) = \mathbf{x}^T M \mathbf{x}$ (M is symmetric), then $\nabla f(\mathbf{x}) = 2M\mathbf{x}$
- If $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$, then $\nabla f(\mathbf{x}) = \mathbf{c}$

Therefore, the FONC for the least squares problem is

$$X^T X \boldsymbol{\beta} = X^T \mathbf{y}$$

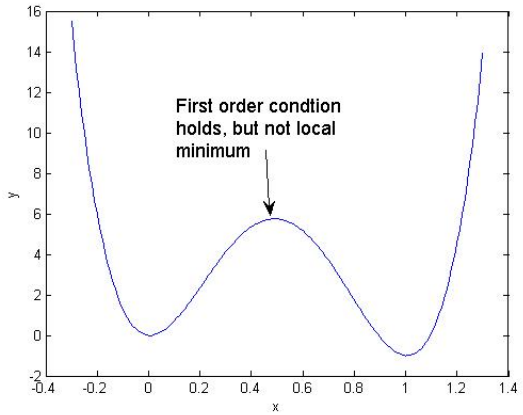
Solving this equation gives candidates for local minimizer.

FONC is Not Sufficient

In the example $f(x) = 100x^2(1-x)^2 - x$. The FONC is

$$f'(x) = 400x^3 - 600x^2 + 200x - 1 = 0$$

with solutions $x_1 = 0.01032$, $x_2 = 0.47997$ and $x_3 = 1.00972$.



We see that FONC is not sufficient (neccessary condition)

- In fact, each local maximum also satisfies the FONC.
- Or it could be neither a local minimum nor maximum (x^3)

Second-Order Necessary Condition

(Not sufficient ?)

Consider the Taylor expansion again but to the 2nd order (assuming f is second-order differentiable):

$$f(\mathbf{x}+\alpha\mathbf{d}) = f(\mathbf{x}) + \alpha\nabla f(\mathbf{x})^T\mathbf{d} + \frac{1}{2}\alpha^2\mathbf{d}^T\nabla^2 f(\mathbf{x})\mathbf{d} + o(\alpha^2)$$

($\nabla f(x) = 0$)

When the first-order necessary condition holds, we have

$$f(\mathbf{x}+\alpha\mathbf{d}) = f(\mathbf{x}) + \frac{1}{2}\alpha^2\mathbf{d}^T\nabla^2 f(\mathbf{x})\mathbf{d} + o(\alpha^2)$$

should be non-negative.

In order for $\mathbf{x}$ to be a local minimizer, we also need $\mathbf{d}^T\nabla^2 f(\mathbf{x})\mathbf{d}$ to be nonnegative for any $\mathbf{d}$.

Second-Order Necessary Condition (SONC)

Theorem (Second-Order Necessary Condition)

If $\mathbf{x}^*$ is a local minimizer of $f(\cdot)$ for an unconstrained problem, then we must have

1. $\nabla f(\mathbf{x}^*) = 0$;
2. *For all* $\mathbf{d}$, $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} \geq 0$.

Definition

We call a (symmetric) matrix A positive semi-definite (PSD) if and only if for all $\mathbf{x}$, $\mathbf{x}^T A\mathbf{x} \geq 0$.

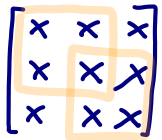
Remark

Therefore, the second-order necessary condition requires the Hessian matrix at $\mathbf{x}^*$ is PSD. In the one-dimensional case, this is equivalent to that the second derivative at $\mathbf{x}^*$ is nonnegative.

Positive Semidefinite Matrices

Here are some useful facts about PSD matrices:

- We usually only talk about PSD for symmetric matrix. If a matrix A is not symmetric, we use $\frac{1}{2}(A + A^T)$ to define the PSD properties (because $\mathbf{x}^T A\mathbf{x} = \frac{1}{2}\mathbf{x}^T(A + A^T)\mathbf{x}$)
- A symmetric matrix is PSD if and only if all the eigenvalues are nonnegative.
- A symmetric matrix is PSD if and only if all the principal submatrices have nonnegative determinants



- For any matrix A, $A^T A$ is a (symmetric) PSD matrix

If A is PSD, we call $-A$ a negative semi-definite matrix.

Example Continued

In the example $f(x) = 100x^2(1-x)^2 - x$, the second-order condition is

$$6x^2 - 6x + 1 \geq 0$$

Only $x_1 = 0.01032$ and $x_3 = 1.00972$ satisfy the condition. But $x_2 = 0.47997$ does not (thus x_2 is not a local minimizer)

In the example of least squares problem, we have the following fact:

- If $f(\mathbf{x}) = \mathbf{x}^T M\mathbf{x}$ (M is symmetric), then $\nabla^2 f(\mathbf{x}) = 2M$

Therefore, the Hessian matrix in that problem is $2X^TX$, which is always a PSD matrix. Therefore, the SONC always holds.

SONC is Not Sufficient

However, even both the first- and second-order necessary conditions hold, it still can't guarantee a local minimum.

Consider $f(x) = x^3$ at 0.

- $f'(0) = f''(0) = 0$, thus FONC and SONC hold.
- 0 is not a local minimum

By modifying the SONC, we can get a sufficient condition.

Second-Order Sufficient Condition (SOSC)

Theorem

Let f be second-order continuously differentiable. If $\mathbf{x}^*$ satisfies:

1. $\nabla f(\mathbf{x}^*) = 0$;
2. *For all* $\mathbf{d} \neq 0$, $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} > 0$.

$f(x+\alpha d) = f(x) + \alpha \nabla f(x) d + \alpha^2 d^T \nabla^2 f(x) d + o(\alpha^2)$

if strictly > 0

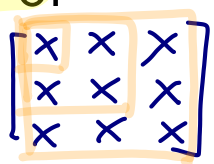
must $\exists$ small α, $f(x+\alpha d) > f(x)$

Then $\mathbf{x}^*$ is a local minimum of f for the unconstrained problem.

Definition

We call a (symmetric) matrix A positive definite (PD) if and only if for all $\mathbf{x} \neq 0$, $\mathbf{x}^T A \mathbf{x} > 0$.

- PD matrix must be PSD (thus PD is a stronger notion)
- A symmetric matrix is PD if and only if all its eigenvalues are positive
- A symmetric matrix is PD if and only if the determinants of all leading principal submatrices are positive
- If A is PD, then we call $-A$ a negative definite matrix.



Proof

The proof is again by Taylor expansion.

When $\nabla^2 f(\mathbf{x}^*)$ is positive definite, we have $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} > c||\mathbf{d}||^2$ where $c > 0$ is the smallest eigenvalue of $\nabla^2 f(\mathbf{x}^*)$.

Thus for any $\mathbf{x}^*$ that satisfies the second-order sufficient condition, for any $\mathbf{d}$, and small enough α

$$f(\mathbf{x}^* + \alpha\mathbf{d}) = f(\mathbf{x}^*) + \frac{1}{2}\alpha^2 \mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} + o(\alpha^2) > f(\mathbf{x}^*)$$

which implies that $\mathbf{x}^*$ is a local minimizer.

For Maximization Problems

Our conditions are derived for minimization problems.

- For maximization problems, it is just the opposite direction

Theorem (FONC for Maximization)

If $\mathbf{x}^*$ is a local maximizer of $f(\cdot)$ for the unconstrained problem, then we must have $\nabla f(\mathbf{x}^*) = 0$.

Theorem (SONC for Maximization)

If $\mathbf{x}^*$ is a local maximizer of $f(\cdot)$ for an unconstrained problem, then we must have 1) $\nabla f(\mathbf{x}^*) = 0$; 2) $\nabla^2 f(\mathbf{x}^*)$ is negative semi-definite

Theorem (SOSC for Maximization)

Let f be second-order differentiable. If $\mathbf{x}^*$ satisfies 1) $\nabla f(\mathbf{x}^*) = 0$; 2) $\nabla^2 f(\mathbf{x}^*)$ is negative definite, then $\mathbf{x}^*$ is a local maximizer.

Optimality Conditions

We have learned for unconstrained problems:

- First-order necessary condition
- Second-order necessary condition
- Second-order sufficient condition

In many cases, we can often use these conditions to identify local optimal solutions.

- Use FONC and SONC to identify candidates, and then use the sufficient condition to verify.
- Or if a problem only has one solution that satisfies FONC, and one can reason that the problem must have a finite optimal solution, then that point must be (global) optimal

Example

In the example $f(x) = 100x^2(1-x)^2 - x$, x_1 and x_3 all satisfy second-order sufficient conditions ($f''(x) > 0$), thus are local minimum.

In the least squares problem, if $X^T X$ is positive definite, then the solution β solved from FONC

$$X^T X \beta = X^T \mathbf{y}$$

satisfies the second-order sufficient conditions and must be optimal.

Example

Find the minimum of

$$f(x, y, z) = 2x^2 + 2y^2 + z^2 - 2xy - 2xz - 6y + 7$$

We first look at the FONC. We have

$$\nabla f(x, y, z) = (4x - 2y - 2z, 4y - 2x - 6, 2z - 2x)$$

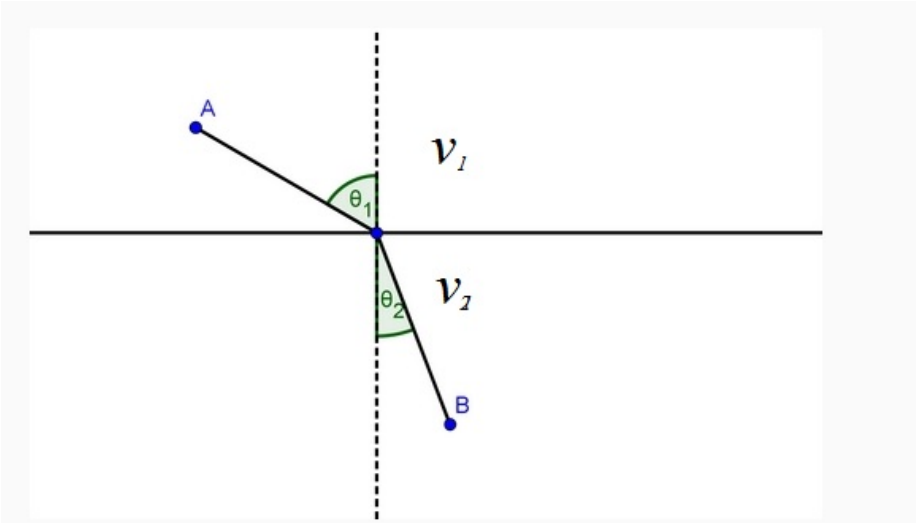
Solve $\nabla f(x, y, z) = 0$, get $x = y = z = 3$.

Now we consider the second-order sufficient condition.

$$\nabla^2 f = \begin{bmatrix} 4 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 2 \end{bmatrix}$$

The eigenvalues of this matrix are all positive. Therefore, $\nabla^2 f$ is positive definite and thus (3,3,3) is a local minimum (in fact it is also the global minimum)

Example: Reflection Principle of Lights



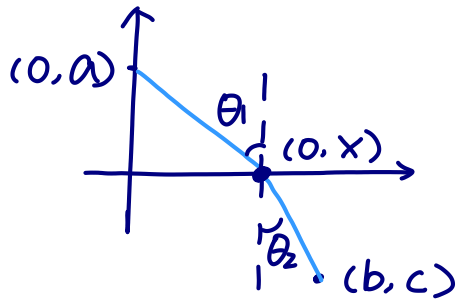
Light travels across two medium which separate at $y = 0$. It travels from $A = (0, a)$ to $B = (b, c)$. The speed in medium 1 is v_1 and the speed in medium 2 is v_2. The basic principle is that light will travel in shortest time.

- What path would the light take?

Example Continued

Suppose it crosses the two medium at $(x, 0)$. Then x must minimize

$$\frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{(x-b)^2 + c^2}}{v_2}$$



The first-order condition is:

$$\frac{1}{v_1} \frac{x}{\sqrt{x^2 + a^2}} + \frac{1}{v_2} \frac{x-b}{\sqrt{(x-b)^2 + c^2}} = 0$$

Therefore

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{x}{\sqrt{x^2 + a^2}} \Big/ \frac{b-x}{\sqrt{(x-b)^2 + c^2}} = \frac{v_1}{v_2}$$

- Heron's reflection principle of light

Next...

We only studied unconstrained problem and local minimum.

- What about constrained problems?
- Global minimum?

We will talk about these in the next lectures.